Introduction

Quasi-Poisson regression is the simplest fix for overdispersion in count data: keep the Poisson mean structure $\log \mu = X^\top \beta$, but introduce a free dispersion parameter ϕ that scales the variance to $\phi\mu$ rather than the Poisson-mandated μ. Coefficient estimates are identical to Poisson; only the standard errors change, multiplied by $\sqrt{\phi}$. It is the workhorse "quick fix" when a Poisson dispersion check fails by a moderate amount and you do not need a likelihood-based comparison.

Prerequisites

A working understanding of Poisson regression, generalised linear models, and the concept of overdispersion.

Theory

Quasi-Poisson uses quasi-likelihood: it specifies only the first two moments $\mathbb{E}[Y] = \mu$ and $\mathrm{Var}(Y) = \phi\mu$, without committing to a particular distribution. The dispersion parameter is estimated by Pearson chi-squared divided by residual degrees of freedom. Coefficient estimates are unchanged from Poisson; standard errors are inflated by $\sqrt{\hat{\phi}}$, and confidence intervals widen accordingly.

Unlike the negative binomial — which has a genuine likelihood and AIC — quasi-Poisson is a quasi-likelihood model: no AIC, no BIC, no likelihood-ratio tests in the usual sense. F-type tests via `anova(fit, test = "F")` are appropriate for comparing nested quasi-Poisson models.

Assumptions

Non-negative integer outcomes, the variance-mean relationship $\mathrm{Var}(Y) = \phi\mu$ (linear in the mean), and independent observations. For quadratic variance-mean (typical of strongly overdispersed counts), negative binomial is more appropriate.

R Implementation

```r
set.seed(2026)
d <- data.frame(x = rnorm(200))
d$y <- rnbinom(200, mu = exp(1 + 0.5 * d$x), size = 2)   # overdispersed

fit_pois <- glm(y ~ x, data = d, family = poisson)
fit_qp   <- glm(y ~ x, data = d, family = quasipoisson)

# Dispersion estimate
summary(fit_qp)$dispersion
# Compare SE
summary(fit_pois)$coefficients
summary(fit_qp)$coefficients
```

Output & Results

The `quasipoisson` family produces the same point estimates as Poisson but with inflated standard errors. The dispersion estimate is in `summary(fit)$dispersion`; comparing the two coefficient summaries shows how much SEs change.

Interpretation

A reporting sentence: "Quasi-Poisson regression estimated a dispersion of $\hat{\phi} = 2.8$, indicating substantial overdispersion; standard errors were inflated by $\sqrt{2.8} = 1.67$ relative to Poisson, giving a 95 % confidence interval for the rate ratio per unit x of 1.30 to 2.00 (Poisson misleadingly gave 1.45 to 1.85)." Always quote the dispersion alongside the inflated SEs.

Practical Tips

- Quasi-Poisson is acceptable for estimation and confidence intervals, but it lacks a proper likelihood — no AIC, BIC, or LRTs in the usual sense.
- For formal model selection by AIC, switch to negative binomial (`MASS::glm.nb` or `glmmTMB(family = nbinom2)`); the additional dispersion parameter is genuinely estimated.
- Quasi-Poisson assumes linear variance-mean ($\text{Var}(Y) = \phi\mu$); NB assumes quadratic ($\text{Var}(Y) = \mu + \mu^2/\theta$). Inspect Pearson residuals against fitted values to choose between them.
- For large dispersion ($\hat{\phi} > 3$), neither quasi-Poisson nor NB may suffice; consider zero-inflated or mixture models.
- Use F-tests (`anova(fit, test = "F")`) for nested quasi-Poisson model comparisons; LRTs are not available.
- For longitudinal or clustered counts, `glmmTMB(family = nbinom2)` is preferable to quasi-Poisson because it provides random effects and a proper likelihood.

R Packages Used

Base R `glm()` with `family = quasipoisson`; `MASS::glm.nb()` for negative binomial as the recommended alternative; `glmmTMB` for mixed-effects count models with proper likelihoods; `DHARMa` for residual diagnostics across count-model families.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI